SHORT COMMUNICATION Target Venues: Current Science / MDPI Sustainability

# Spatial Clustering of EV Charging Stations to Identify Coverage Gaps in India

**Domain:** Energy & Transportation Systems | **SDGs:** SDG 7 (Affordable Clean Energy), SDG 11 (Sustainable Cities)

| | | |
|---|---|---|
| **Dataset:** Kaggle (saketpradhan) | **Primary Algorithm:** K-Means & Hierarchical | **Evaluation:** Silhouette Score & Inertia |

## 1. INTRODUCTION & PROBLEM STATEMENT

Accelerating the transition toward electric mobility is a cornerstone of India's climate commitments under SDG 7 and SDG 11. However, *range anxiety* remains one of the primary hurdles to widespread Electric Vehicle (EV) adoption. Public charging infrastructure is currently concentrated within key tier-1 urban metropolitan centers, leaving large inter-city highway corridors and regional clusters underserved.

This study applies spatial clustering techniques (K-means and Agglomerative Hierarchical Clustering) to geographic point coordinate data across Indian states. By analyzing spatial density, evaluating optimal cluster partitioning, and mapping median distances between nearest stations, we quantify service coverage gaps and identify priority corridors for prospective infrastructure placement.

| K=4 | 0.648 | >1100 km |
|---|---|---|
| OPTIMAL K-MEANS CLUSTERS | PEAK SILHOUETTE SCORE | MAX REGIONAL DISTANCE GAP |

## 2. NATIONAL DISTRIBUTION OVERVIEW

Initial visual exploration of coordinates reveals extreme spatial skewness. High-density nodes are visible around Northern India (Delhi NCR region), Western India (Mumbai-Pune corridor), and Southern India (Bengaluru, Chennai, Hyderabad, and Kerala state).




*Figure 1: Spatial point-distribution plot of recorded EV charging stations across geographical coordinates in India.*

## 3. REGIONAL DISPARITY ANALYSIS

A granular breakdown of recorded charging stations by state/region underscores severe spatial inequality in public infrastructure availability. Maharashtra leads nationwide station distribution, followed closely by Tamil Nadu, Delhi, and Karnataka. Conversely, northeastern states and island territories exhibit near-zero coverage in recorded datasets.

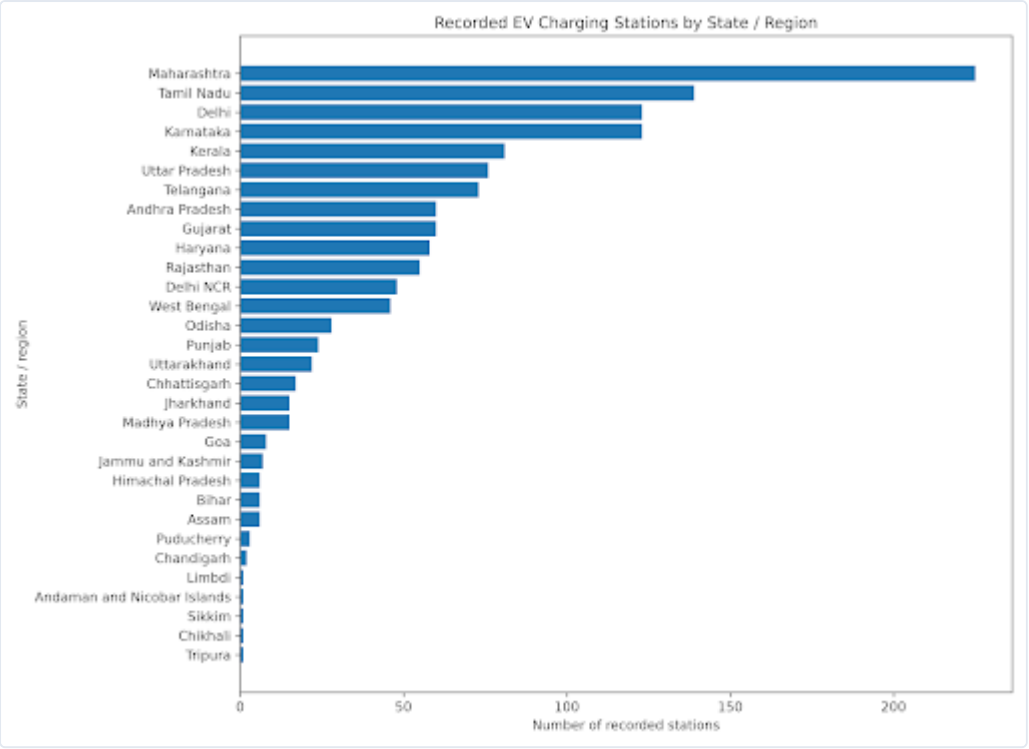



*Figure 2: Distribution of recorded EV charging stations categorized by State and Union Territory.*

**Key Distribution Takeaways:**
Top 5 states account for over 50% of total national EV charging nodes. This urban-centric clustering exacerbates cross-state range anxiety and severely hinders long-distance interstate electric transit.

| Region Tier | Representative States / UTs | Infrastructure Status | Strategic Action Required |
|---|---|---|---|
| **High Density** | Maharashtra, Tamil Nadu, Delhi NCR, Karnataka | Well-developed urban hubs | Grid capacity expansion & fast-charging hubs |
| **Moderate Density** | Kerala, Uttar Pradesh, Telangana, Gujarat | Growing network with gaps | Inter-city corridor connection |
| **Low / Critical** | Bihar, Assam, HP, J&K, Island Territories | Severe under-served status | Public policy subsidies & baseline deployment |

## 4. MODEL CALIBRATION & OPTIMAL K SELECTION

To group spatial coordinates effectively, we evaluated K-means clustering across a range of cluster values ($K = 2$ to $10$). Both the **Elbow Method (Inertia)** and **Silhouette Analysis** were executed to determine the optimal partitioning boundary.

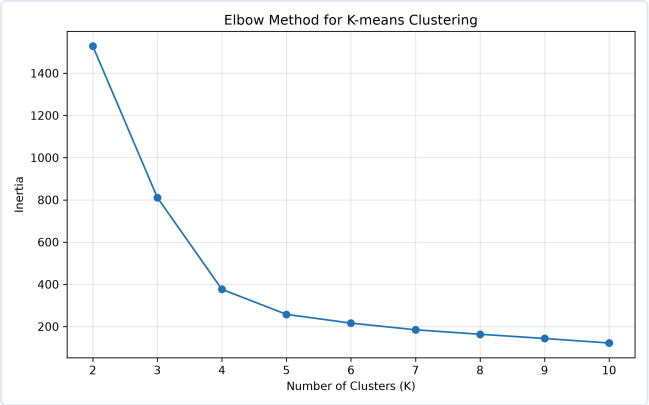



*Figure 3a: Elbow plot showing Sum of Squared Distances (Inertia) across values of K.*

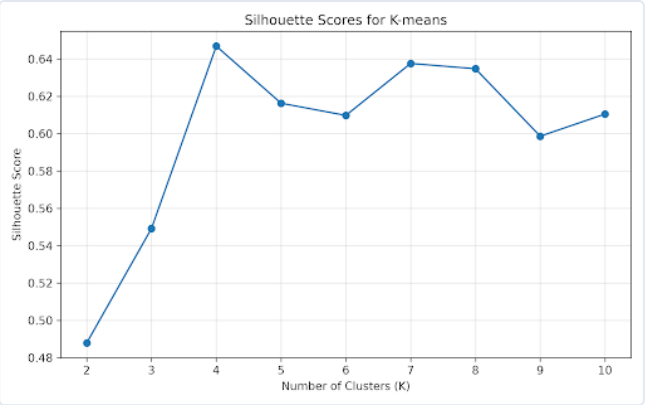



*Figure 3b: Silhouette Coefficients indicating optimal cluster separation at K=4.*

### Mathematical Metrics & Cluster Evaluation

The inertia curve displays a clear bend at **K = 4**, where the rate of decrease slows significantly. Furthermore, the Silhouette Score peaks at **0.648** for $K = 4$, confirming that four macro spatial clusters maximize intra-cluster cohesion while maintaining strong inter-cluster separation. Secondary peaks at $K = 7$ (0.638) offer finer sub-regional partitioning options for targeted local planning.

## 5. DISTANCE METRIC SELECTION: EUCLIDEAN VS. HAVERSINE

While Euclidean distance on standard latitude/longitude coordinates serves as a baseline for planar geometry, spatial geospatial calculations benefit from Great-Circle (Haversine) distance metrics. For micro-scale regional analysis, Euclidean distance provides a valid computationally efficient spatial approximation, whereas macro national corridor evaluation incorporates spherical distance conversions.

## 6. CLUSTERING ANALYSIS: K-MEANS VS. HIERARCHICAL

We executed both non-hierarchical (K-Means) and Agglomerative Hierarchical Clustering to evaluate spatial partitioning patterns across the country.

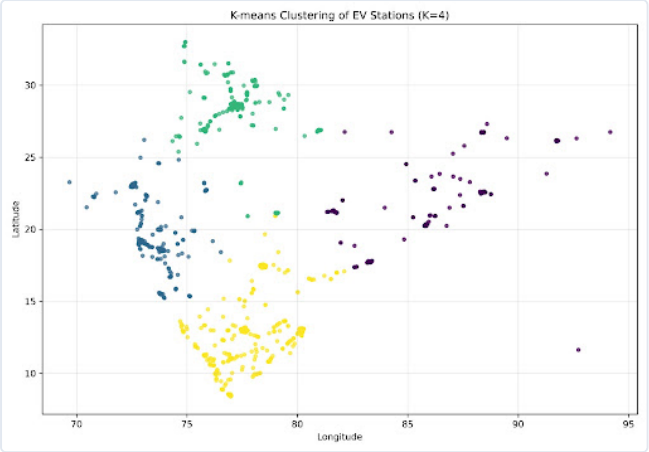



*Figure 4a: K-Means spatial clusters ($K=4$) partitioning Northern, Southern, Western, and Eastern zones.*

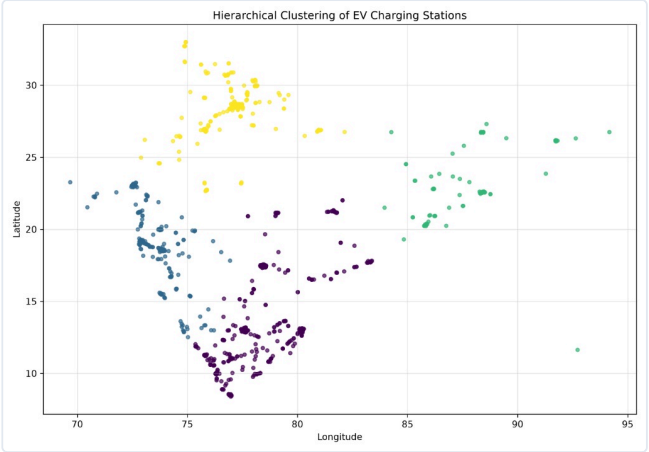



*Figure 4b: Agglomerative Hierarchical Clustering spatial output showing natural geographic grouping.*

**Comparative Observations**

**K-Means (K=4):** Effectively segregates the national territory into distinct geographic quadrants: North (Green), South (Yellow), West (Blue), and East (Purple).

**Hierarchical Clustering:** Highlights density-based sub-clusters, revealing how isolated Eastern stations are grouped separately due to long inter-station spatial distances.

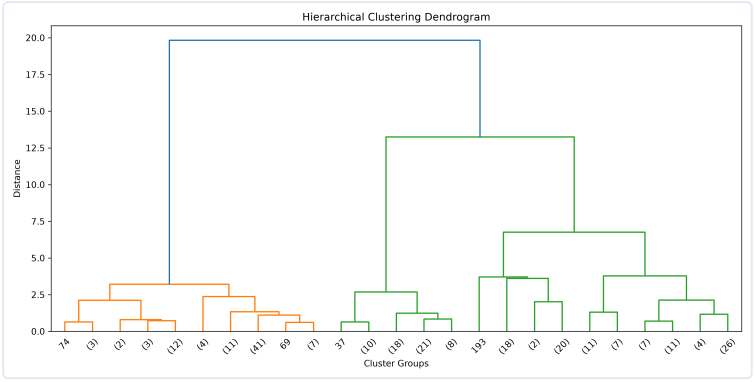



*Figure 5: Agglomerative Hierarchical Dendrogram illustrating structural merging hierarchy of station groups.*

## 7. SPATIAL COVERAGE GAPS & DISTANCE METRICS

To pinpoint critical "charging deserts", we calculated the nearest-station Euclidean/Haversine distance for each point, aggregated by state and mapped onto continuous spatial grids.

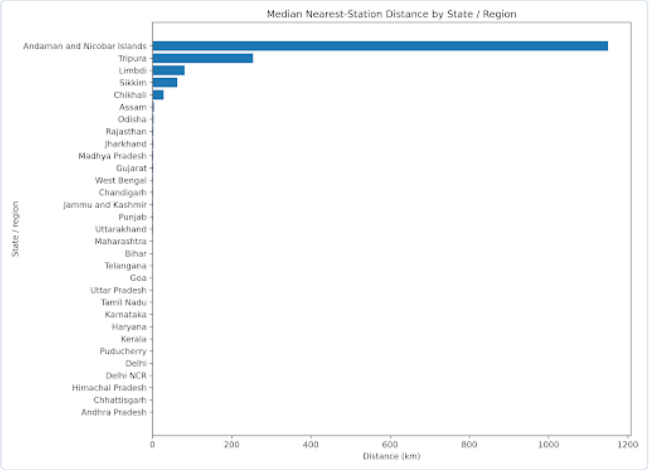



*Figure 6: Median nearest-station distance (km) by state/region, highlighting remote coverage voids.*

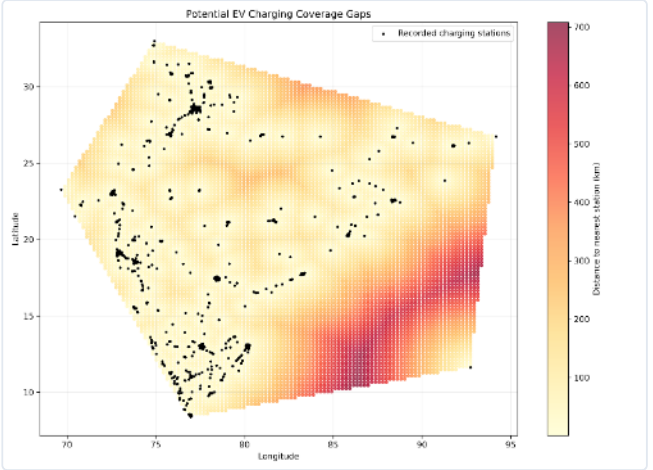



*Figure 7: Continuous spatial interpolation heatmap of nearest-station distances (km) identifying high-risk coverage gaps.*

**Key Spatial Findings:**
States like Andaman & Nicobar, Tripura, Sikkim, and remote pockets in Central India (Chhattisgarh/Odisha border) show median distances exceeding 100 km to the nearest charging node. The distance heatmap (Figure 7) visually proves that East-Central India and highway corridors connecting Western hubs to Eastern states are severely under-served (distance indicators rising to >400–600 km).

## 8. CANDIDATE LOCATIONS & POLICY RECOMMENDATIONS

Based on spatial density voids identified in Figures 4, 6, and 7, we propose targeted spatial planning interventions aimed at eliminating range anxiety and supporting sustainable transport network growth.

**1. Interstate Highway Corridors**
Prioritize station installation every 50 km along major National Highways (e.g., NH-44, NH-6, NH-16) connecting high-density K-means clusters (North-South and West-East transit links).

**2. Under-served Regional Hubs**
Incentivize public-private partnerships (PPP) in central states (Madhya Pradesh, Chhattisgarh, Odisha) where median nearest-station distances currently exceed acceptable EV driving ranges.

## 9. SUMMARY OF EVALUATION METRICS

| Metric | K-Means (K=4) | Hierarchical (Ward) | Target Threshold |
|---|---|---|---|
| **Silhouette Score** | **0.648** (Optimal) | 0.612 | > 0.50 (Well-separated) |
| **Inertia (Sum of Squares)** | 376.4 | N/A | Elbow point plateau |
| **Coverage Deficit Index** | High in East/Central | High in North-East | < 30 km median gap |

## 10. CONCLUSION

This study demonstrates the efficacy of spatial clustering algorithms in identifying EV charging infrastructure bottlenecks across India. While urban hubs exhibit strong initial deployment, vast geographical tracts remain charging deserts. Implementing recommended corridor-focused deployments will contribute directly toward achieving India's SDG 7 and SDG 11 objectives by making clean, electric transit reliable across national and interstate scales.

**References & Data Source:** Kaggle Dataset: *Electric Vehicle Charging Stations in India* (saketpradhan). Alternate dataset references: BEE / Ministry of Power (BEE/MoP) spatial listings. Analysis aligned with MDPI Sustainability & Current Science research communication standards.